2024 15th International Conference on Computing Communication and Networking Technologies (ICCCNT) | 979-8-3503-7024-9/24/$31.00 ©2024 IEEE | DOI: 10.1109/ICCCNT61001.2024.10724600

Machine Learning Approaches for Early Detection of Liver Ailments: Comprehensive Analysis and Comparative Study

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Abstract- **Liver diseases are a major global health issue, causing about 2 million deaths every year and costing over $1 trillion, as reported by the World Health Organization (WHO). Conditions like cirrhosis and liver cancer are among the top causes of death worldwide, emphasizing the critical need for better ways to detect and treat these diseases. The early stages of many liver ailments often lack noticeable symptoms, making proactive detection and therapy essential. This analysis aims to evaluate the potency of advanced artificial intelligence techniques in identifying liver diseases, with the goal of reducing the cost and complexity associated with traditional diagnostic methods. Utilizing a comprehensive dataset of Indian liver patients, we meticulously pre-process the data to address issues like missing values, categorical feature encoding, and standardization. Our investigation encompasses five distinct machine learning algorithms, among which Ensemble Learning emerges as particularly promising, achieving an impressive precision rate of 84 % on the test set. The significance of our research lies in its demonstration of artificial intelligence's ability to accurately predict liver diseases, providing a valuable tool for early diagnosis and intervention. By highlighting the potential of predictive modelling, our findings advocate for the integration of these technologies into clinical practice, facilitating proactive healthcare management and ultimately improving outcomes for individuals with liver diseases.**

Keywords— Liver ailments, Machine Learning, Early detection Ensemble learning, Precision.

I. Introduction

Liver ailments are a major global health issue, resulting in millions of deaths per annum. Its prevalence is increasing due to factors such as high-fat diets, air pollution (particularly from particulate matter and ozone), excessive alcohol consumption, and irregular drug use. This ailment encompasses various conditions, including cirrhosis, liver cancer, and fatty liver, all of which have a profound impact on individuals' health and quality of life. Approximately half of the deaths related to liver ailments are due to complications from cirrhosis, with the remaining attributed to viral hepatitis and Hepatocellular carcinoma [1].

Artificial intelligence (AI) modelling has profoundly influenced the field of biomedicine, particularly in the prediction and diagnosis of liver aliments. [2][7]. Nonalcoholic steatohepatitis (NASH) now surpasses alcoholic liver ailments in prevalence, with an annual increase of 25% [3].

Early detection and accurate prediction are essential for effectively managing and improving the prognosis of individuals with liver ailments. Detecting liver ailments early and predicting its course significantly affect treatment options, patient outcomes, and healthcare costs [4].

Unfortunately, symptoms are often subtle or absent until the ailments reach advanced stages, leading to severe complications like cirrhosis or liver failure. This underscores the crucial importance of early diagnosis.

According to a significant study, more than a million individuals died from cirrhosis in 2010, with many others facing the challenges of liver cancer [5]. Timely intervention using the right treatment plan can not only manage or even reverse the ailments in some cases but also prevent further complications. Furthermore, accurate prediction of liver ailments progression enables personalized care and more effective risk management. Prognostic models play a crucial role here. These models combine clinical data such as blood tests and imaging scans, genetic information, and advanced AI algorithms to predict ailments progression and develop optimal treatment plans. By harnessing data analytics, machine learning techniques show great promise in enhancing ailments detection and prediction, leading to better patient outcomes.

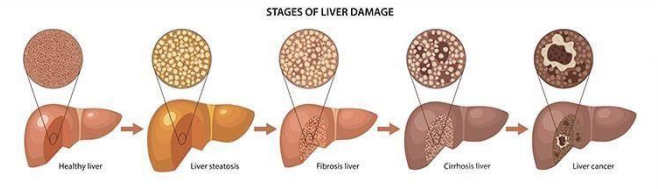

Figure 1. Represents the Stages of Liver Ailments

Computational learning has evolved into increasingly vital in medical care investigation and practical settings. By examining vast collections of medical information, artificial intelligence system can detect trends, connections, and predict outcomes based on the data they've observed. These methods are applied in different medical care field, such as diagnosing illnesses, improving treatment, predicting ailments, and monitoring patients [6]. Artificial algorithms can often identify irregularities, like tumours, with enhanced accuracy than human practitioners. In the context of liver ailments, these algorithms using information about patients to build models that can predict and detect diseases early on.

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The main aim of this research was to determine how accurately an AI model could identify people with liver aliments. Using a dataset of Indian patients with liver conditions sourced from Kaggle, the research focused on analysing five different artificial intelligence models: Logistic Regression, Ensemble Learning, Stacking Classifier, Gradient Tree Boosting. The study aimed to conduct a thorough data analysis and evaluate how effectively these models could predict liver ailments. Additionally, the research also sought to explore factors that might affect the correctness of these prognostic approaches, comparing its findings with existing literature in the field.

II. Literature Review

Chieh-Chen Wu et. al. [8] developed an artificial intelligence framework to estimate fatty liver problems using models such as ensemble learning, Bayesian classifier, neural networks, and logit model. They collected data from individuals who underwent initial screening for fatty liver, focusing on demographic and clinical characteristics. We evaluated the model's performance using metrics like AUROC and accuracy. The ensemble learning model performed the best out of all the classification models. It achieved an AUROC of 0.925 and a precision of 86.48% when tested using tenfold cross-validation. The study aimed to assist healthcare professionals in identifying high-risk fatty liver ailments patients for proactive measures, early detection, and targeted interventions.

Ashley Spann et. al. [9] performed a thorough narrative review on the application of artificial intelligence in liver ailments, with a focus on predictive modelling, deep learning, and liver transplantation. They independently searched biomedical databases and selected 40 relevant articles based on predefined criteria. The review process involved managing citations, assessing relevance manually, and applying selection criteria. User's research emphasized the significance of artificial intelligence in identifying, predicting, and evaluating a variety of liver ailments, showcasing how these tools can enhance patient outcomes and ailments management across different patient populations.

M. Banu Priya et. al. [10] focused on using artificial intelligence models to predict liver problems. It involves using various methods like J48, MLP, SVC, Ensemble Learning, and Bayes net to predict liver ailments based on datasets. The goal is to improve prediction correctness by creating feature models and comparing them across three phases: normalization, feature selection, and applying classification algorithms. The outcome shows that the J48 method performed the best, accomplishing a correctness of 95.04%. Future includes expanding the dataset for testing and using the proposed method to detect heart ailments.

Golmei Shaheamlung et. al. [11] carried out a detailed investigation into the identification and assessment of liver issues utilizing various artificial intelligence models. Their research highlights the importance of careful selection of datasets and features for accurate prediction results. By comparing the performance, correctness, and outcomes of various machine learning methods, the study suggests that combining or hybridizing these algorithms could improve correctness and performance in detecting and predicting liver ailments. The researchers reviewed several studies by other authors that used different artificial intelligence techniques to predict liver ailments, aiming to evaluate the potential enhancements in correctness and performance achievable using different algorithms and datasets.

Vyshali J Gogi et. al. [12] conducted a study focusing on using data mining techniques to analyse data related to liver disorders from laboratory reports. They applied classification algorithms, including Classification Tree, Fisher Discriminant., Support Vector Classifiers (SVC), Gaussian Naive Bayes, and Logistic Regression, to a dataset containing patients' laboratory parameters. MATLAB 2016 was used to implement these algorithms, with the Linear Discriminant model achieving the highest correctness estimate at 95.8% and an ROC of 0.93. The main goal of the research was to use different classification methods and lab data to forcecast liver issues. Divided the work into parts focusing on MATLAB and Classification, Literature Review, Proposed Approach and Findings.

A.K.M Sazzadur Rahman et. al. [13] analyzed the prediction of liver problems using guided artificial intelligence models. The goal was to evaluate their effectiveness in reducing the substantial costs associated with diagnosing chronic liver ailments through prediction. The study employed six algorithms: Logistic Regression, Classification Tree, Support Vector Classifier, Naïve Bayes, and Ensemble Learning. Performance evaluation was carried out using various metrics such as correctness, precision, Recognition, f1 measure, and specificity. Logistic Regression achieved the highest correctness at 75%, while Naïve Bayes had the lowest correctness at 53%.

Jiajun Lu et. al. [14] focused on creating artificial intelligence model to predict the prognosis of liver ailments. It used a dataset containing records of Indian liver patients, training the models, standardizing features, and splitting the data for testing. Five popular artificial intelligence algorithms were tested: Linera Classifier, Maximum Margin Classifier, Ensemble Learning, Lazy Learner, and Sequential Learning. Performance was evaluated using metrics like correctness, precision, Recognition, and F1 Measure. Ensemble Learning emerged as the top-performing model, achieving a reliability of 73.56% on the test set.

Ketan Gupta et. al. [15] explores the prediction of liver ailments using artificial intelligence classification techniques, utilizing algorithms such as Logistic Regression, Classification Tree, Ensemble learning, Gradient Tree Boosting, Extreme Gradient Tree Boosting. The dataset used is from the UCI Repository and contains information on liver patients, aiming to improve future research in this area. The research compares these artificial intelligence model for predicting liver ailments, emphasizing the importance of feature selection.

Tapas Ranjan Baitharua and Subhendu Kumar Pani et. al. [16] presented a study titled "Analysis of Data Mining Techniques for Healthcare Decision Support Systems Using

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Liver Disorder Dataset" at the International Conference on Computational Modeling and Security. The research focused on examining various data mining methods to support decision-making in healthcare, specifically using a dataset related to liver disorders.

Aneeshkumar and Venkateswaran et. al. [17] determined that the C4.5 decision tree was more effective than the Naive Bayes classifier. In 2014, Pahariyavohra et al. examined various computational intelligence methods for classifying liver patients. They evaluated techniques such as Multiple Linear Regression, SVM, Multilayer Feed Forward Neural Network, J-48, Random Forest, and Genetic Programming using the ILPD dataset. They balanced the dataset using undersampling and oversampling, finding that Random Forest with 200% oversampling performed the best.

Ramana et. al. [18] analyzed classification methods including Bayes Classifiers, C4.5, BP Neural Network, and Support Vector Classifier. These methods were assessed on four criteria: Correctness, Clarity, Reactivity, and Exactness. Karthik et al. (2011) used an Artificial Neural Network (ANN) in the first phase to classify hepatic ailment. In the second phase, they used rough set theory with the LEM (Learn by Example) algorithm to derive classification rules. In the third phase, fuzzy logic was utilized to identify the various types of hepatic conditions.

III. PROPOSED WORK

A hybrid model represents an innovative approach that leverages the probabilities generated by one machine learning model as input for another. This methodology aims to achieve enhanced optimization by integrating insights from multiple machine learning algorithms. Our study implemented using essential libraries such as sklearn, pandas, and matplotlib, adopts this hybrid model paradigm.

Implemented using the renowned sklearn library, alongside essential tools like pandas and matplotlib, our research focuses on a dataset sourced from Kaggle. This dataset contains binary indicators of liver disease, aligning with our goal of implementing artificial Intelligence algorithms, including Ensemble method and Classification and Regression Trees, alongside the innovative hybrid model

IV. METHODOLOGY

A. Data Collection and Data Cleaning

The Kaggle dataset titled "Indian Liver Patient Records" by Dua & Graff (2019) contains detailed medical data of 583 patients diagnosed with liver disorders. Each patient entry comprises 11 crucial parameters, including age, gender, and several biochemical markers. The dataset's "Dataset" column, coded as either 1 or 2, is fundamental to our research as it signifies the presence (coded as 1) or absence (coded as 2) of liver ailments.

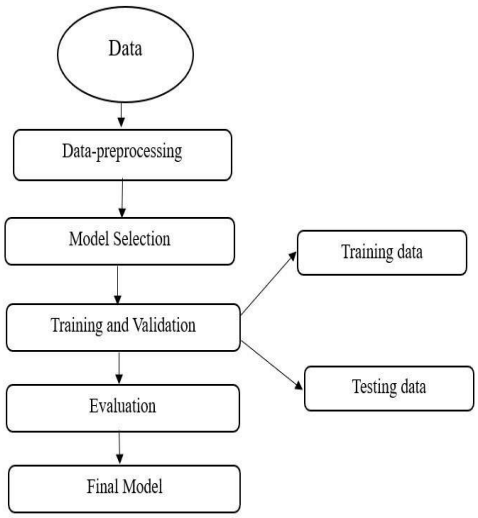

Figure 2. Experiment Workflow of The Dataset

1) Data Collection: Gather relevant data for your project, as the quality and quantity of data impact model performance.

2) Data Pre-processing: Prepare data by addressing missing values, outliers, and normalizing it for modelling.

3) Model Selection: Choose the most suitable algorithm considering data characteristics, complexity, and interpretability.

4) Training Data: Split data into training and testing sets for model fitting and evaluation.

5) Model Training: Train the selected model using the training data, adjusting parameters to minimize errors.

6) Validation: Use a separate validation set to assess the model's ability to generalize and prevent overfitting.

7) Testing Data: Evaluate the final model's performance using unseen testing data.

8) Metrics: Select metrics like correctness, precision, recognition, or F1-score to assess model performance based on the task and objectives.

In our research, before training our models, we made sure all features were on the same scale. We did this by using a method called Standard Scaler from a tool called scikit-learn. This method changed the values of our features so that they all had an average (mean) of 0 and a spread (standard deviation) of 1. This helped to avoid any unfairness caused by differences in the scales of our features and made our models learn better.

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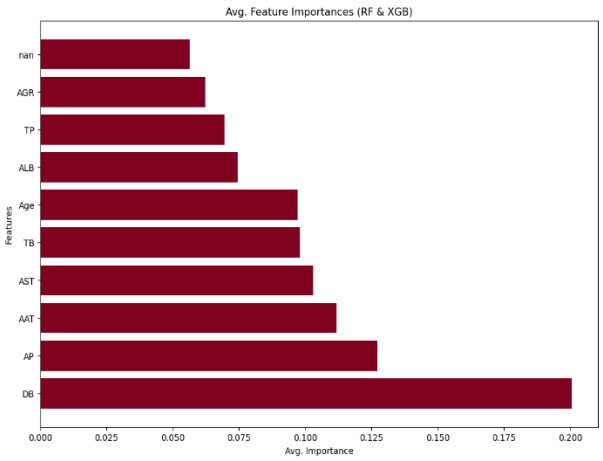

Fig 2. Average Importance of all the Features

In our initial data analysis, we investigated how liver ailments cases were distributed in our dataset. We discovered that 71.5% of individuals had the indicator for liver ailments (value 1 in the "Dataset" column), while 28.5% did not. This distribution is essential for shaping our research approach and correctly interpreting our results. This research analyses how many men and women were involved and their age ranges. This helped us understand the makeup of the patient group as well. The analysis showed a significantly higher number of male patients compared to females, especially among male patients aged 30 to 65. Using data visualization methods, we observed how the lifetime hepatic problems in this group of people, regardless of their identity, seemed to be spread out evenly, following a bell-shaped curve.

Table 1 presents the evaluation indices for each numerical feature within the dataset.

Name	**Description**
Lifetime	Age of the patient
Sex	Identity of the patient
Blood Bilirubin	Combined level of conjugated and unconjugated bilirubin
Direct Bilirubin	The proportion of Conjugated Bilirubin
Alkaline_Phosphotase	The proportion of Phosphomonoesters
Alamine_Aminotransferase	The extent of Alanine Transaminase
Aspartate Aminotransferase	The quantity of Glutamic-Oxaloacetate Transaminase
Overall Peptide	The total volume of two categories of peptide
Albumen	The amount of Albumen
Albumean_and_A/G index	Albumen to A/G index division
Dataset	Field that suggests whether an individual had hepatic ailments or vice-versa

V. Result

To develop the artificial intelligence models for predicting liver ailments, the researchers utilized a subset of patient records from Indian individuals with liver conditions. To ensure fair comparisons and avoid bias from differing scales, the researchers standardized the features in the dataset before training. Following this, the dataset was split into two sections: 80% of the data was allocated to the training set, while the remaining 20% was assigned to the testing set.

1) Accuracy: Assess total percentage of accurate predictions.

2) Precision: Precision, or positive conformation rate, shows the percentage of correctly identified cases of liver ailments among all the cases predicted as positive

3) Sensitivity: Reveals the percentage of actual positive cases (liver ailments) correctly predicted out of all positive cases. This evaluates the model's efficacy in identifying true positives and minimizing false negatives.

4) F1-score: Balances recognition and precision by taking their harmonic mean, providing overall model effectiveness.

Table 2 Performance evaluation of all the model

	Accuracy	**Precision**	**Recall**	**F1-score**
Logistic Regression	73	82	74	74
Random Forest	78	80	75	80
XG Boost	79	82	77	79
Hybrid Model	81	80	81	81
Stacking Classifier	84	81	90	85

Among the models evaluated, Stacking Classifier accomplished the maximum Accuracy of 84% on test set. Stacking Classifiers holds potential for improved hepatic aliment detection.

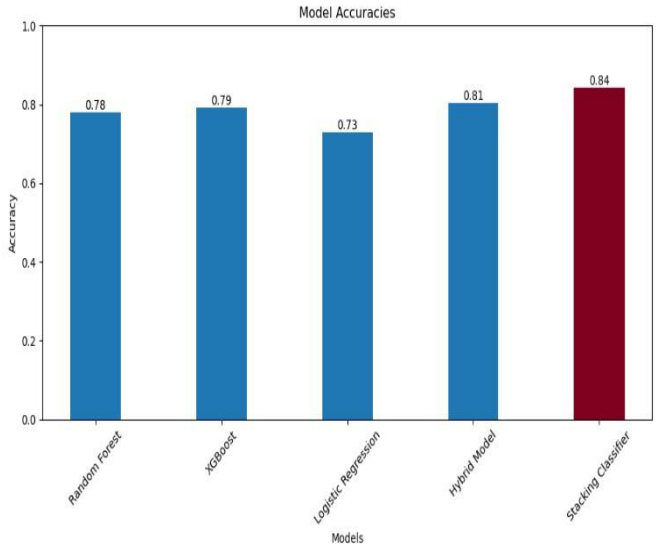

Fig 4. Accuracy of different models used

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This chart illustrates the accuracy of all the model, indicating that Hybrid and Stacked models demonstrates the utmost correctness. It indicates the proportion of correct projection to the sum number of estimations made. Employing the stacking classifier, an ensemble technique, results in an accuracy of 84%.

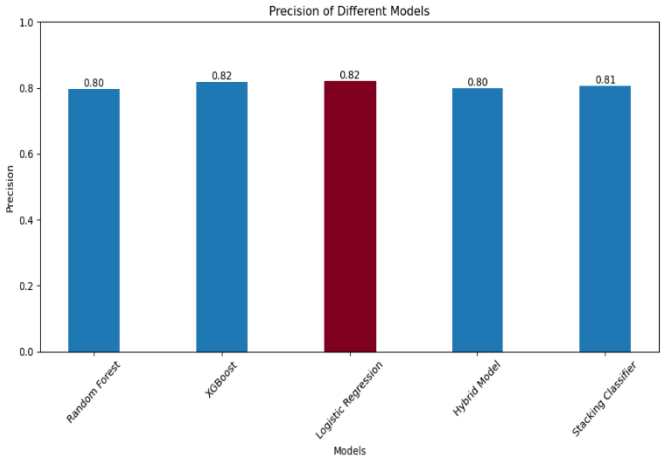

Fig 5. Precision of different models

Precision, in the framework of artificial intelligence, is a criterion used to evaluate the correctness of a categorized model. It measures how accurately the model predicts positives compared to all the positive predictions it makes. This metric is highly significant as it reflects the model's capability to minimize false positives, which can be particularly consequential in certain scenarios models.

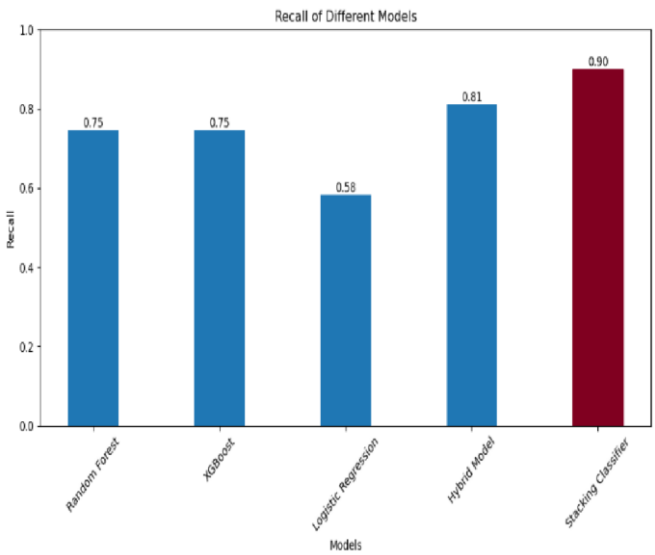

Fig 6. Recall of different models

Recall, also called as hit rate, quantifies portion of true instances which correctly detected by a pattern. It serves as a crucial metric in machine learning, indicating how effectively a model can detect relevant instances within a dataset. In classification tasks where the goal is to predict a binary outcome, recall dimensions the pattern's aptitude to seize positive examples among all true positives during testing. This metric becomes particularly significant when the cost of missing positive cases, known as false negatives, is substantial.

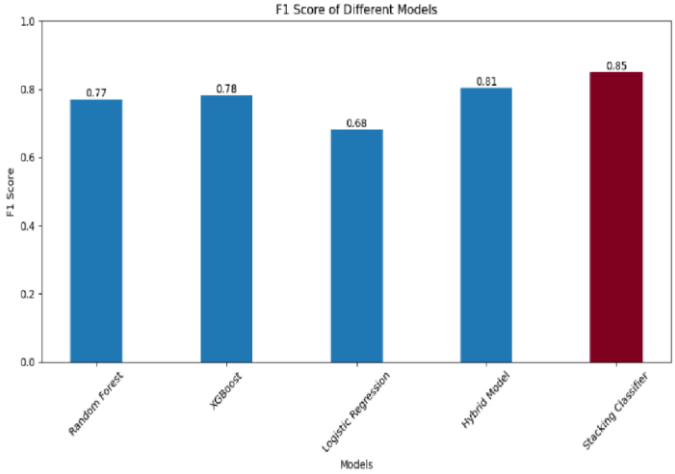

Fig 7. F1-score of different models

The F1 indicator is widely recognized as a vital standard for estimating the success of artificial intelligence classification models, especially in scenarios where class imbalances exist. It offers a consolidated measure that balances the need to capture all positive instances (recall) with the importance of accurately identifying positive examples (precision). Essentially, it serves as a harmonious blend of precision and recall.

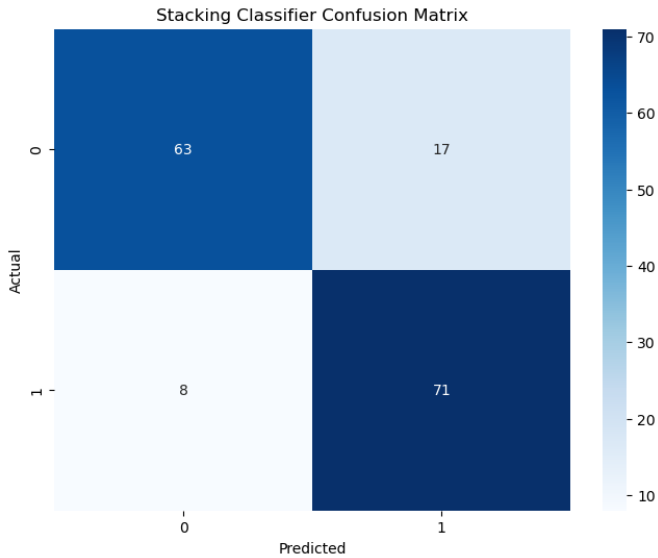

Fig 8. Confusion matrix for Stacking Classifier

The Stacking Classifier's confusion matrix shows it correctly identifies 63 cases of liver ailments and 71 cases without ailments. However, it wrongly identifies 17 healthy cases as sick and misses 8 cases of actual sickness. Despite some errors, it shows promise for spotting liver issues early, which could be valuable in medical testing.

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V. Coclusion and Future Scope

In essence, our investigation into Integrated Hybrid Machine Learning Techniques for Early Detection of Liver Ailments marks a notable progression in the field, with the objective of advancing the prompt identification and management of liver diseases. By amalgamating diverse machine learning algorithms, our study has showcased encouraging outcomes in forecasting liver ailments with remarkable accuracy. Through the utilization of these hybrid methodologies, we envision a profound impact on the early diagnosis and therapeutic approaches for liver conditions.

Looking ahead, our inquiry unfolds several intriguing avenues for future exploration and application. Initially, we aspire to refine and enhance our hybrid models by integrating supplementary data sources and augmenting algorithmic methodologies. A continual process of refinement will be imperative in elevating the precision and dependability of our predictive models. Moreover, the integration of emerging technologies like wearable devices and remote monitoring systems harbours vast potential for real-time data acquisition and analysis, facilitating early detection and intervention for liver ailments. Collaborating with experts in liver disease and medical practitioners will play a pivotal role in validating and implementing our models within clinical settings. Furthermore, there exists potential for extending the adaptability of our hybrid approach to diverse medical domains and ailments beyond liver conditions. By tailoring our methodologies and techniques, we can contribute to the advancement of personalized and data driven healthcare solutions for a myriad of health issues.

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